

INT 453: Digital Multimedia

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Lecture 6

Huffman Coding

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books

- Fundamentals of Multimedia.
- Multimedia Technologies & applications for the 21st century.

Data Compression

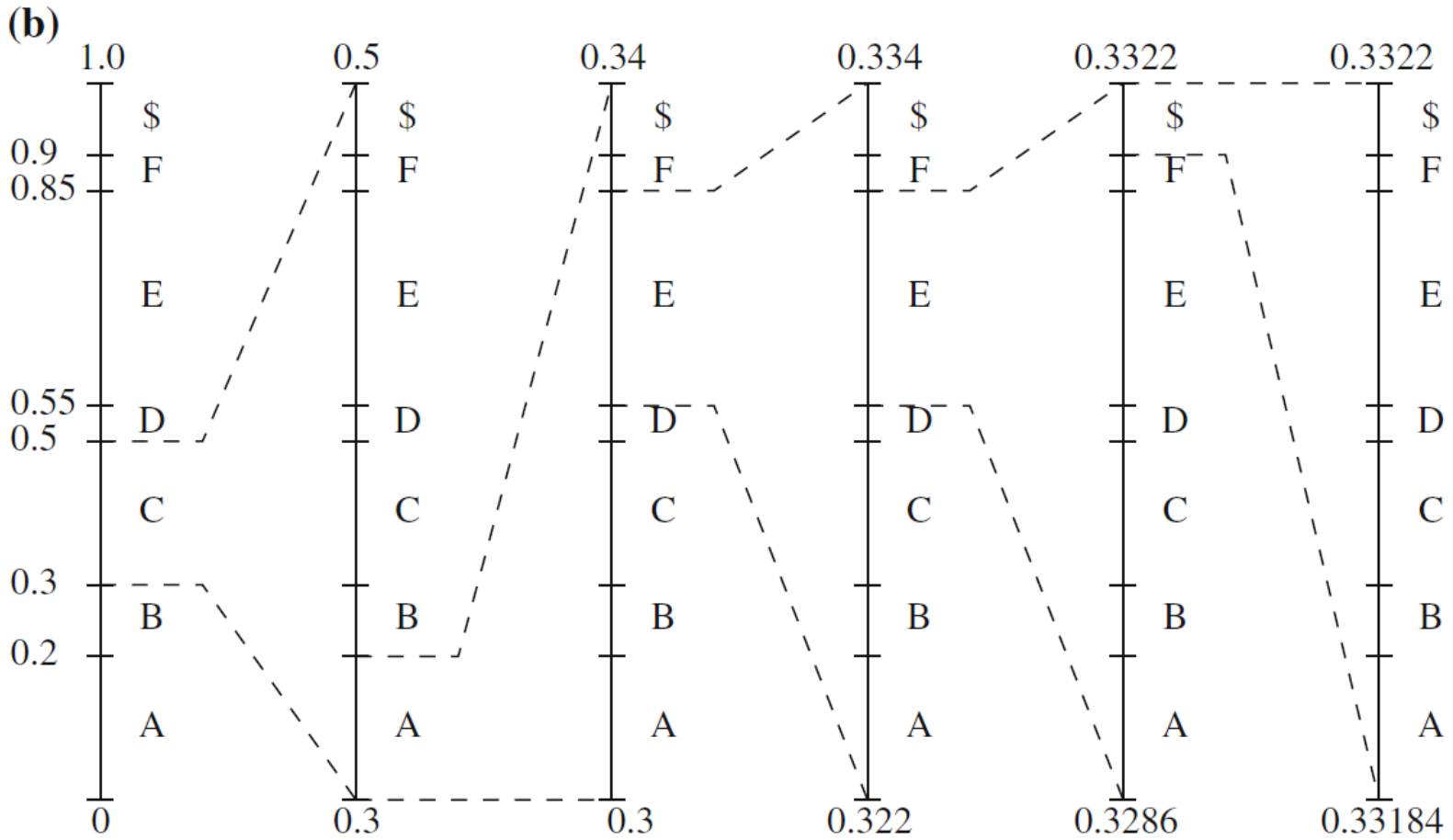
Encoding Example
Ex. 2 (home work)

Encode the String **: CAEE$**

Using the following table

Symbol	Probability	Range	Range_low	Range_high
A	0.2	[0,0.2)	0	0.2
B	0.1	[0.2,0.3)	0.2	0.3
C	0.2	[0.3,0.5)	0.3	0.5
D	0.05	[0.5,0.55)	0.5	0.55
E	0.3	[0.55,0.85)	0.55	0.85
F	0.05	[0.85,0.9)	0.85	0.9
$	0.1	[0.9,1.0)	0.9	1.0

Encoding Example
Ex. 2



Arithematic Decoding Algorithm

```
BEGIN
   get binary code and convert to decimal value = value(code);
   Do
        {
          find a symbol s so that
               Range_low(s) <= value < Range_high(s);
          output s;
          low = Rang_low(s);
          high = Range_high(s);
          range = high - low;
          value = [value - low] / range;
        }
   Until symbol s is a terminator
END
```

Decoding Example 1

Arithmetic decoding example

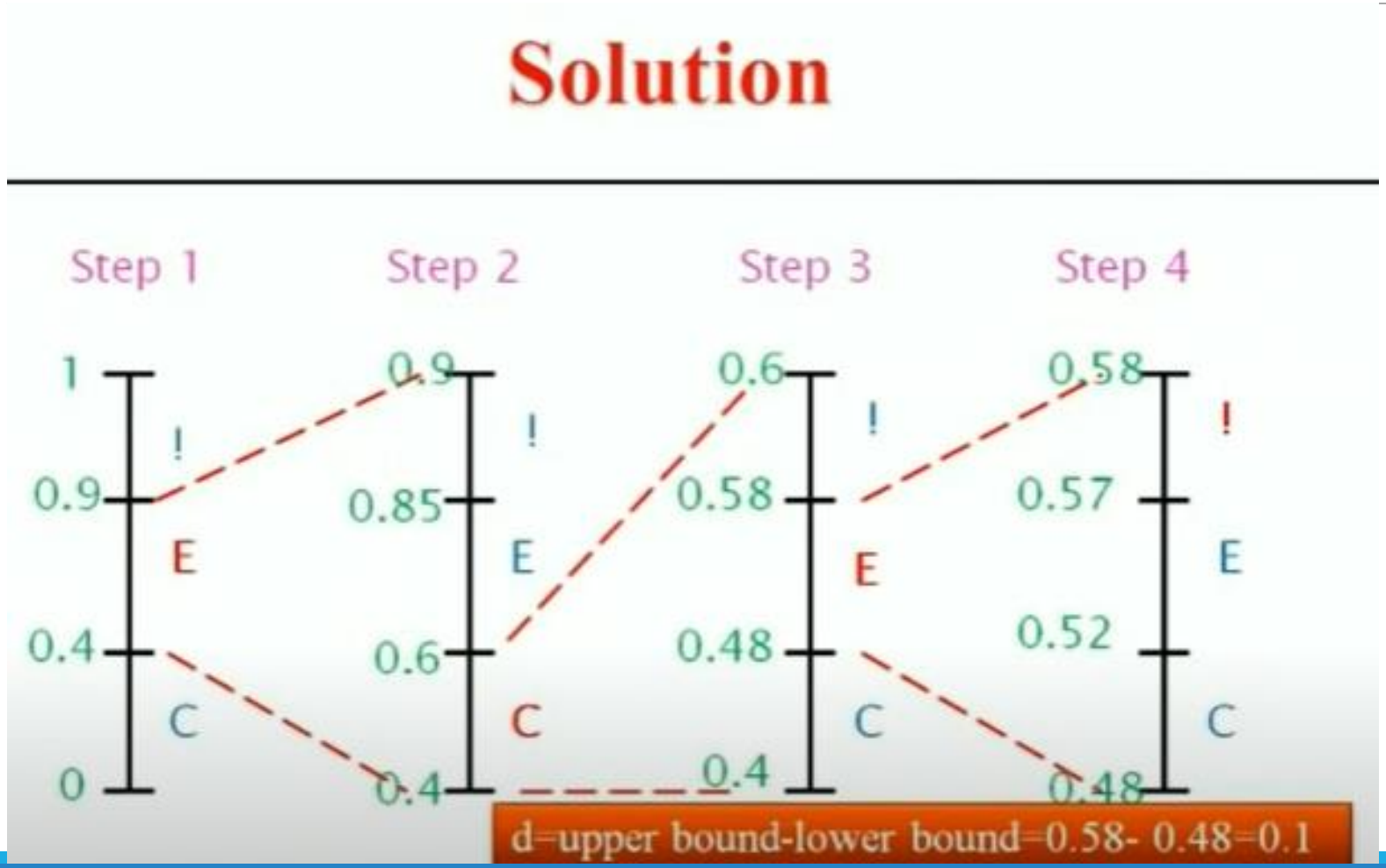
Decode the message 0.572 given the coding model

Symbol	!	C	E
Probability	0.1	0.4	0.5

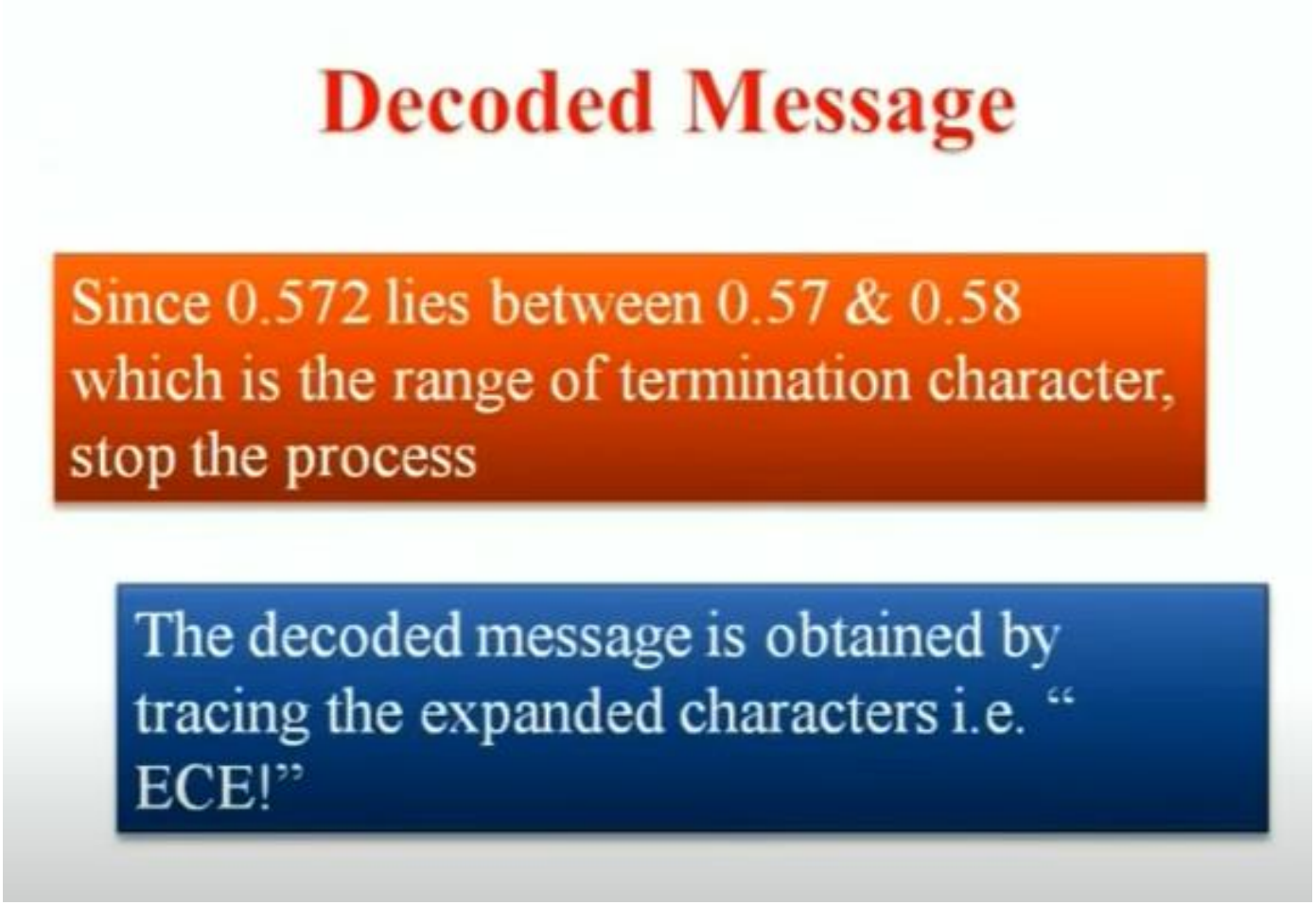
Symbol	!	C	E
Probability	0.1	0.4	0.5

Symbol	Probability	Range	Range_low	Range_high
A	0.2	[0,0.2)	0	0.2
B	0.1	[0.2,0.3)	0.2	0.3
C	0.2	[0.3,0.5)	0.3	0.5
D	0.05	[0.5,0.55)	0.5	0.55
E	0.3	[0.55,0.85)	0.55	0.85
F	0.05	[0.85,0.9)	0.85	0.9
$	0.1	[0.9,1.0)	0.9	1.0

Decoding Example 1 (cont.,)



Decoding Example (cont.,)



Arithmetic Decoding Ex. 2

Arithmetic decoding example

Decode the message 0.33203125 given the coding model

Symbol	Probability	Range	Range_low	Range_high
A	0.2	[0,0.2)	0	0.2
B	0.1	[0.2,0.3)	0.2	0.3
C	0.2	[0.3,0.5)	0.3	0.5
D	0.05	[0.5,0.55)	0.5	0.55
E	0.3	[0.55,0.85)	0.55	0.85
F	0.05	[0.85,0.9)	0.85	0.9
$	0.1	[0.9,1.0)	0.9	1.0

Arithmetic Decoding Ex. 2

Note that each time we use the law:

New_value=(value-low range) / difference

The decoding process for the above example.

1- Initially, value = 0.33203125.

- Since Range_low*(C)* = 0.3 ≡ 0.33203125 < 0.5 = Range_high*(C)*, then the first output symbol is C.
- This yields *value* = [0.33203125 − 0.3]/0.2 = 0.16015625,
- So, it determines that the second symbol is A.

Eventually, *value* is 0.953125, which falls in the range [0.9, 1.0*)* of the terminator $.

symbol	prob	low range	high range	d (difference)	value	symbol located in which interval?	new value=(value-low(symbol located in)) / diff(symbol located in)
A	0.2	0.0	0.2	0.2	0.3320313	C	0.16015625
B	0.1	0.2	0.30	0.1	0.1601563	A	0.80078125
C	0.2	0.3	0.50	0.2	0.8007812	E	0.83593750
D	0.05	0.5	0.55	0.1	0.8359375	E	0.95312500
E	0.3	0.6	0.85	0.3	0.9531250	$	**stop here** Eventually, *value* is 0.953125, which falls in the range [0.9, 1.0) of the terminator $.
F	0.05	0.9	0.90	0.1			
$	0.1	0.9	1.00	0.1			
						the decoded string is CAEE$	

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Huffman Coding

It is a popular technique for removing coding redundancy.

The result after Huffman coding is variable length code, where the code words are unequal length.

Huffman coding yields the smallest possible number of bits per gray level value.

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1. **Build a Huffman Tree :**
2. 1. Combine the two lowest probability leaf nodes into a new node.
   2. Replace the two leaf nodes by the new node and sort the nodes according to the new probability values.
   3. Continue the steps (a) and (b) until we get a single node with probability value 1.0. We will call this node as **root**

Huffman Coding

Consider the following string with each symbol's frequency, Encode data using huffman encoding diagram

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Symbol	e	i	o	p	b	c
Frequency	57	51	33	20	12	3

Huffman Coding

Consider the following string with each symbol's frequency, Encode data using huffman encoding diagram

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Symbol	e	i	o	p	b	c
Frequency	57	51	33	20	12	3
Probability	0.32	0.29	0.19	0.11	0.07	0.02

Huffman Coding

e	.32	11	.32	11	.32	11	.39	0	.61	1
i	.29	10	.29	10	.29	10	.32	11	.39	0
o	.19	00	.19	00	.20	01	.29	10		
p	.11	011	.11	011	.19	00				
b	.07	0101	.09	010						
c	.02	0100								

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https://www.geeksforgeeks.org/image-compression-using-

Huffman Coding - Example

Consider the 8-bit gray image shown below. Use Huffman coding technique for eliminating coding redundancy in this image:

119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

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Solution

Gray level	Histogram	Probability
107	3	0.1875
119	8	0.5
123	2	0.125
168	3	0.1875

Let's sort the probabilities in an ascending order

Add the two least probabilities

And then resort and add less

Huffman Coding - Example

119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

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Solution

0.5

0.1875

0.1875

0.125

These two nodes are origintaed from the 0 code, so they will have 01 and 00 respectively

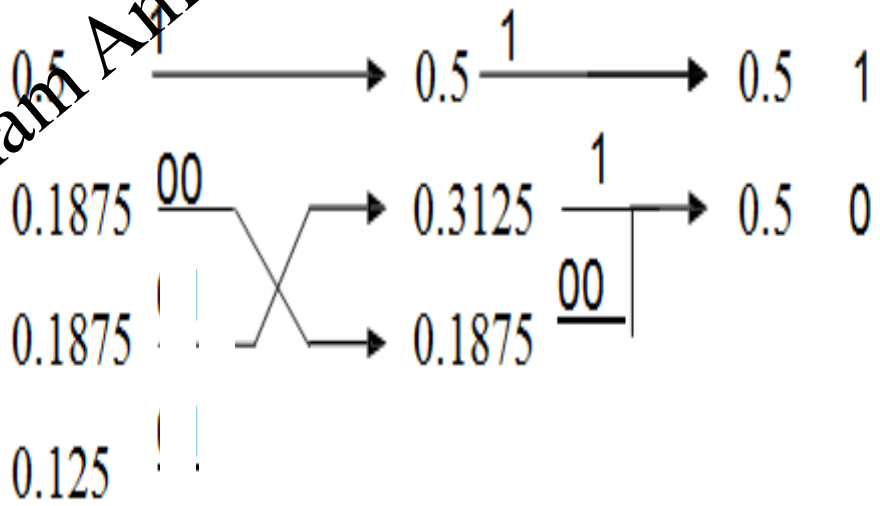
To see how to label each arrow with code
Start from the last nodes

Huffman Coding - Example

Consider the 8-bit gray image shown below. Use Huffman coding technique for eliminating coding redundancy in this image:

119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

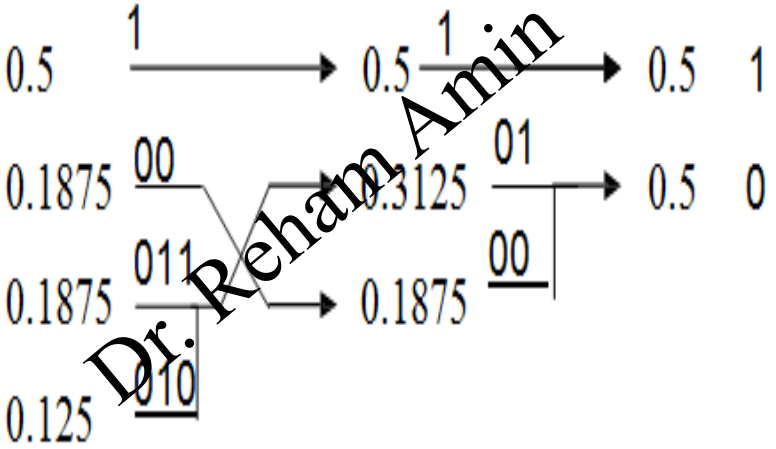
Solution



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https://www.youtube.com/watch?v=acEaM
-Mfw

Huffman Coding - Example



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119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

Solution

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

Lookup table:

Gray level	Probability (Frequency / 16)	Frequency	Code	Code frequency
119	0.5	8	1	0.5*1=0.5
168	0.1875	3	00	0.1875*2=0.375
107	0.1875	3	011	0.1875*3=0.5625
123	0.125	2	010	0.125*3=0.375
			Average length of the encoded pixels	1.8125 bits/pixel

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Huffman Coding - Example

Consider the 8-bit gray image shown below.

Use Huffman coding technique for eliminating coding redundancy in this image:

119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

Gray level	Probability	Code
119	0.5	1
168	0.1875	00
107	0.1875	011
123	0.125	010

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Solution

Replace each gray level in the original image with the associated code in the look up table

1	010	00	1
010	1	00	00
1	1	011	1
011	011	1	1

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Encode data{119, 168, 107} using huffman encoding diagram 100011

Thus 77% of the **original 8-bit fixed-length** intensity representation was removed as coding redundancy.

1	010	00	1
010	1	00	00
1	1	011	1
011	011	1	1

Consider the 8-bit gray image shown below.

Use Huffman coding technique for eliminating coding redundancy in this image:

119	123	168	119
123	119	168	168
119	119	107	119
107	**107**	**119**	**119**

Solution

the total number of bits required to represent the gray levels of the compressed image $(M*N*L_{avg})$ = 4*4*1.8125=29 bits is

29-bits:

10101011010110000011110011

(column by column).

Whereas the original (uncompressed) image requires 4*4*8 =128 bits.

Compression ratio CR = 128 / 29 ≈ 4.4

How much you save? =1- 1/CR =1 – ¼.4=0.77

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Huffman coding example 3

Huffman Coding - Example

Consider the 5-bit gray image (10 * 10) with the following frequency, using Huffman coding encode it??

Symbol (like Gray level)	Frequency
A1	10
A2	40
A3	6
A4	10
A5	4
A6	30

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Huffman Coding - Example

Consider the 5-bit gray image (10 * 10) with the following frequency:

Solution

Symbol (like Gray level)	Frequency	Probability
A1	10	10/100=0.1
A2	40	40/100=0.4
A3	6	6/100=0.06
A4	10	10/100=0.1
A5	4	4/100=0.04
A6	30	30/100=0.3

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Huffman Coding - Example

Let’s sort probabilities in ascending order

Solution

Symbol (like Gray level)	Probability
A2	0.4
A6	0.3
A4	0.1
A1	0.1
A3	0.06
A5	0.04

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Huffman Coding - Example

Let’s sort probabilities in ascending order

Solution

Symbols (like intensity levels)	Probabilities (sorted)	Source Reduction (do till two values are left) (Maintain in sorted order here as well)	
a2	0.4		
a6	0.3		
a1	0.1		
a4	0.1		
a3	0.06		
a5	0.04		

Huffman Coding - Example

Let's sort probabilities in ascending order

Solution

Symbols (like intensity levels)	Probabilities (sorted)	Source Reduction (do till two values are left) (Maintain in sorted order here as well)			
		1	2	3	4
a2	0.4 1	0.4 1	0.4 1	0.4 1	0.6 0
a6	0.3 00	0.3 00	0.3 00	0.3 00	0.4 1
a1	0.1 011	0.1 011	0.2 010	0.3 01	
a4	0.1 0100	0.1 0100	0.1 011		
a3	0.06 01010	0.1 0101			
a5	0.04 01011				

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Huffman Coding - Example

Let's sort probabilities in ascending order

Symbol (like Gray level)	Probability	Code	#bits/code	Avg bits
A2	0.4	1	1	0.4
A6	0.3	00	2	0.6
A4	0.1	011	3	0.3
A1	0.1	0100	4	0.4
A3	0.06	01010	5	0.3
A5	0.04	01011	5	0.2

L avg=0.4+0.6+0.3+0.4+0.3+0.2= 2.2

Solution

Symbols (like intensity levels)	Probabilities (sorted)	Source Reduction (do till two values are left) (Maintain in sorted order here as well)			
		1	2	3	4
a2	0.4 1	0.4 1	0.4 1	0.4 1	0.6 0
a6	0.3 00	0.3 00	0.3 00	0.3 00	0.4 1
a1	0.1 011	0.1 011	0.2 010	0.3 01	
a4	0.1 0100	0.1 0100	0.1 011		
a3	0.06 01010	0.1 0101			
a5	0.04 01011				

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Huffman Coding - Example

Let's sort probabilities in ascending order

Solution

Symbol (like Gray level)	Probability	Code	#bits/code	Avg bits
A2	0.4	1	1	0.4
A6	0.3	00	2	0.6
A4	0.1	011	3	0.3
A1	0.1	0100	4	0.4
A3	0.06	01010	5	0.3
A5	0.04	01011	5	0.2

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The compression rate achieved= 10*10*5/10*10*2.2= 2.27

Review Questions

Generate the code word for the string “BCAB” with the probability A →0.4, B →0.3, C →0.2, D →0.1.

Encode the String**: CAEE$** Using the following table.

Symbol	Probability
A	0.2
B	0.1
C	0.2
D	0.05
E	0.3
F	0.05
$	0.1

Review Questions

Decode the message 0.752 given the coding model

Symbol	m	C	A	E
Probability	0.1	0.3	0.2	0.4

Decode the message 0.624 given the coding model

Symbol	S	B	A	E
Probability	0.1	0.2	0.3	0.4

Assignment

- What are the different types of Arithmetic Coding?
- What are the limitations of using Arithmetic Coding?
- How to find the compression ratio using Arithmetic Coding.

Questions

Thank You